

Project – High Level Design On Agriculture System Triage Agent

Manufacturing Operational Insights Report

Course Name: Agentic AI

Institution Name: Mediacaps University – Datagami Skill Based Course

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Problem Statement & Objectives

1. Problem Statement

Farmers often face delays in receiving agricultural advice related to crop diseases, irrigation, fertilizers, weather issues, and government schemes. Traditional support systems rely heavily on manual intervention, leading to slow response times and inefficient case prioritization.

The problem is to design a **production-grade reactive Agriculture Support Triage Agent** that:

- Automatically reads farmer queries
- Classifies intent
- Detects urgency
- Extracts key agricultural entities
- Generates helpful advisory responses
- Escalates high-risk cases to experts

2. Project Objectives

- Accept farmer queries in natural language
- Perform intelligent intent classification
- Extract crop, symptoms, and environmental context
- Evaluate severity and urgency
- Generate actionable step-by-step advice
- Escalate critical cases to agriculture experts
- Store conversations for learning and analytic

3. Scope of the Project

The project focuses on:

- AI-based text processing
- Intent classification
- Named Entity Recognition (NER)
- Urgency detection
- LLM-based response generation

- Routing to human experts
- Session and state management
- Caching for performance improvement

Proposed Solution

The Agriculture Support Triage Agent processes farmer messages step-by-step through an intelligent pipeline.

Step 1 — Message Input

Farmers send messages through:

Mobile app

SMS/WhatsApp chatbot

Web portal

Example message:

“My tomato plants have yellow leaves and spots after rain. What should I do?”

Step 2 — Intent Classification

The system identifies the purpose of the message: **Possible Intent**

Meaning

Disease Issue	Crop infected
Irrigation Problem	Water shortage/excess
Fertilizer Advice	Nutrient guidance
Weather Concern	Rain/frost warning
Scheme Inquiry	Government benefits

Step 3 — Urgency Detection

The agent decides priority level: **Actions**

Urgency

High	Immediate expert escalation
Medium	AI advice + queue review
Low	Automatic response

Example:

Pest outbreak → High urgency

Fertilizer suggestion → Medium urgency

General information → Low urgency

Step 4 — Information Extraction (NER)

- Extracts important details from farmer message
- Identifies crop name
- Detects symptoms described
- recognizes location of farm
- Finds time or environmental condition (after rain, night, morning, etc.)
- Estimates severity of problem (mild / spreading / severe)

Step 5 — Response Generation

- Uses extracted information and detected intent
- Predicts likely cause of the issue
- Suggests immediate practical solution
- Provides preventive measures
- Advises contacting agriculture officer if problem continues

Step 6 — Routing Decision

- Severe or high-risk disease → forward to agriculture expert immediately
- Moderate issue → send AI advice and keep under monitoring
- General or informational query → automatic reply to farmer

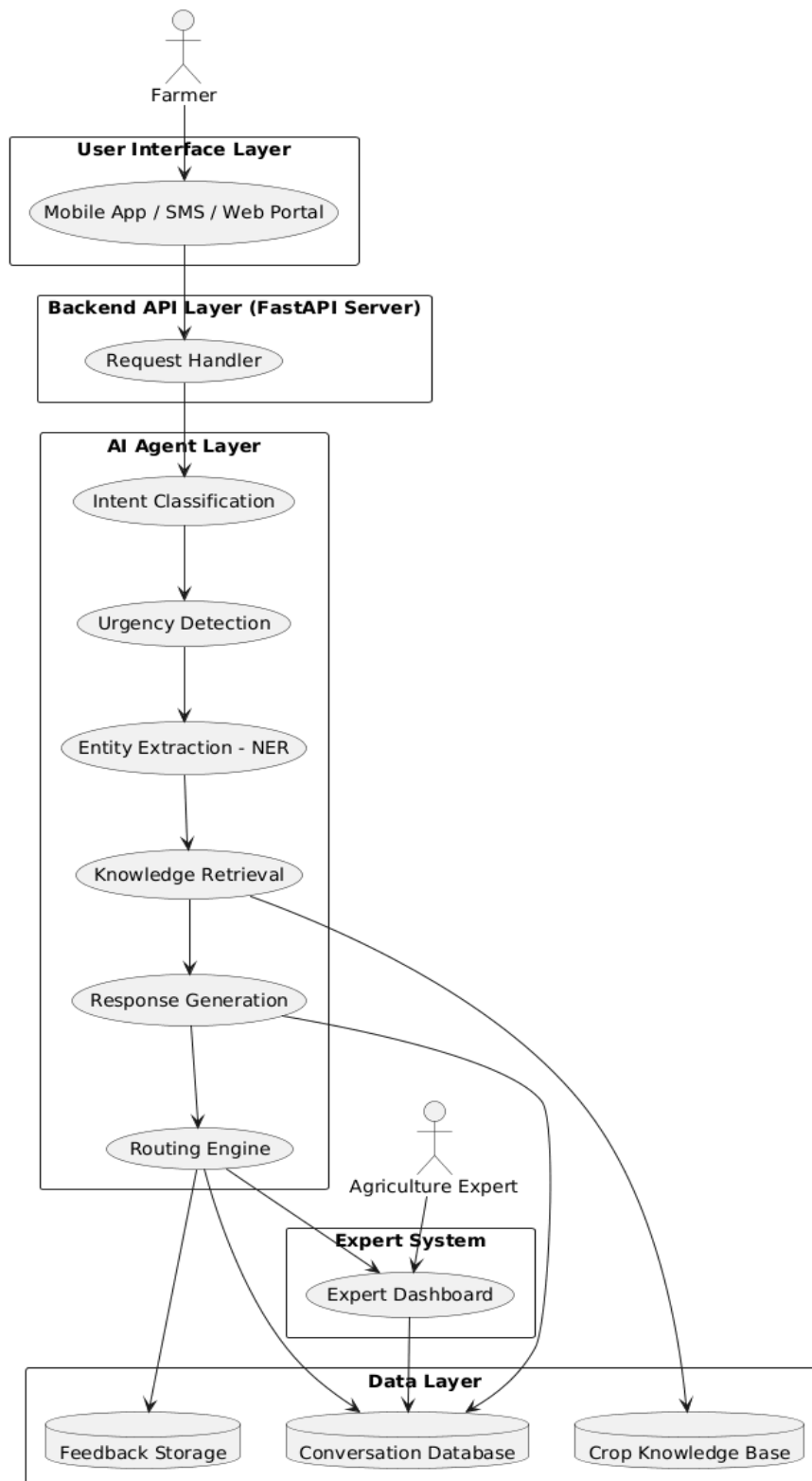
1. Key features

- Real-time intent classification
- Named Entity Recognition (crop, symptoms, time, location)
- Urgency detection (Low / Medium / High)
- LLM-powered advisory generation
- Expert escalation system
- Conversation history storage
- Session management
- Caching using Redis
- Role-based authentication
- Secure API communication (HTTPS)

2. Overall Architecture / Workflow

Architecture Layers:

1. User Interface Layer
 - Mobile App / Web UI / SMS
2. Backend API Layer (FastAPI)
 - Request handling
 - Session management
 - Data storage
3. AI Processing Layer (CrewAI Agents)
 - Query Understanding Agent
 - NER Extraction Agent
 - Urgency Assessment Agent
 - Advisory Generation Agent
4. Data Layer
 - SQLite Database
 - Knowledge Base
 - Feedback Database

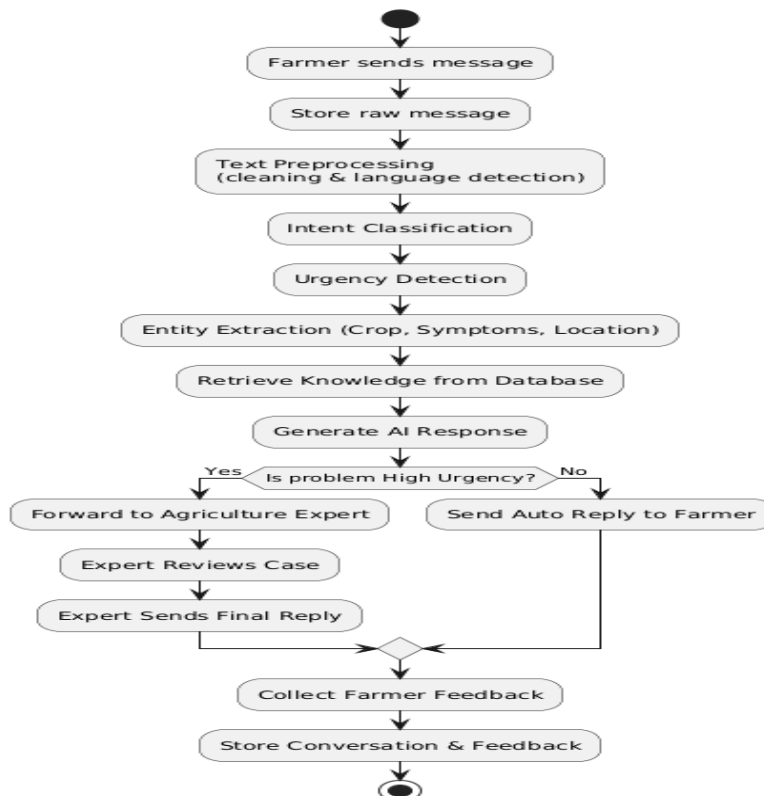


Workflow Summary:

The message moves step-by-step through the system.

- Farmer submits message
- System stores raw message
- Language detection and cleaning performed
- AI identifies intent
- AI determines urgency
- System extracts crop, symptoms, location and time
- AI generates suggested solution
- Routing decision taken
- If urgent → send to human expert
- If normal → auto reply sent
- Feedback collected from farmer
- Data stored for learning

Agriculture Support Triage Agent - Process Flow



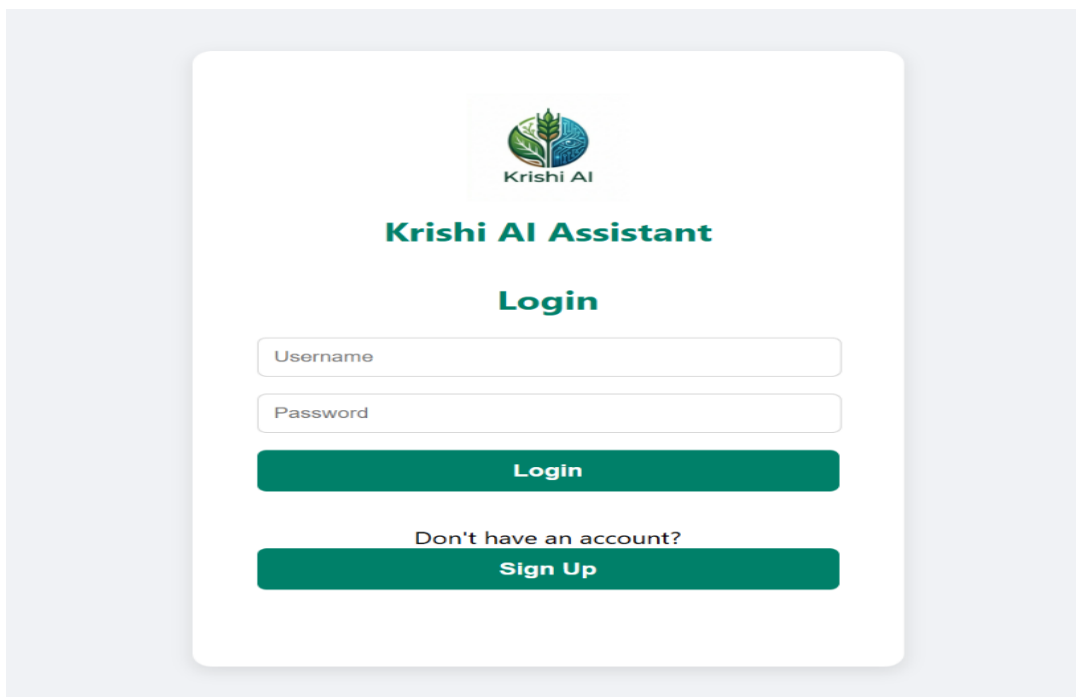
3. Tools & Technologies Used

- Python – Core programming language
- FastAPI – Backend API framework
- CrewAI – AI agent orchestration
- LLM API – Natural language understanding & response generation
- SQLite – Database storage
- React – Web UI
- JWT – Secure Authentication

Results & Output

1. Screenshots / outputs

Login page-



User interface-


Krishi AI Assistant
Logout

Send

Urgency /ner classification/intent finding


Krishi AI Assistant
Logout

Karnataka

corn is being rusted

AI-Detected Urgency: Medium

Crop Name: Corn (likely susceptible to rust, which is a common issue in warm and humid climates)

Urgency: Medium. Corn rust is a serious disease in Karnataka's climate, where temperatures and humidity can foster infection. Immediate action is necessary within 1-2 weeks to prevent significant yield loss.

Location/Plant Part Affected: The rust infection may be observed on the lower leaf area of the corn plant. In Karnataka, rust is more prevalent on the lower leaves due to the humid microclimate in these regions.

Time/Growth Stage: Rust infection often occurs during the silking and tasseling growth stages, typically around 6-8 weeks after planting when temperatures range from 18-30°C (64-86°F). In a subtropical region like Karnataka, the infection may appear during May and June.

Solution:

Send

Response generated-


Krishi AI Assistant
Logout

Solution:
To address the issue successfully, follow these steps:

- Scouting and monitoring:** Regularly inspect your corn crop for rust infection, examining both top and bottom leaves. Monitor temperature and humidity levels using a thermometer and hygrometer to assess the disease's potential spread.
- Adjust cultivation practices:**
 - * Avoid dense planting to minimize humidity buildup.
 - * Improve air circulation by planting rows in a north-south direction to allow air to penetrate between them.
 - * Remove weeds regularly to reduce humidity levels.
- Integrated pest management (IPM):** Apply fungicides if rust coverage is extensive. Rotate fungicides periodically to avoid fungicide resistance. Use systemic, multi-site, or demethylation inhibitor (DMI) fungicides. Apply fungicides at recommended rates at least 10-14 days apart, according to the label instructions for each product. Some recommended fungicides for corn include pyraclostrobin (Headline, Cabrio, and Quash) and tebuconazole.
- Foliar application:** Apply products with the active ingredient chlorothalonil (e.g., Daconil), azoxystrobin (e.g., Heritage), or pyraclostrobin (e.g., Cabrio) at 3-4 leaf collar stages to minimize rust infestation.
- Crop rotation:** Rotate corn with crops like legumes, sorghum, or pearl millet that help reduce the disease's severity by breaking disease cycles.
- Resistant varieties:** If rust is a recurring issue, consider using disease-resistant corn hybrids. Some corn varieties have natural resistance to rust. Growers may also choose varieties with resistance to other common corn diseases.

Send

2. Reports / models

- *Intent classification*
- *NER extraction*
- *Urgency prediction*
- *Generated advisory*

Example:

Input:

"crop is being rusted"

Output:

- *Intent: Crop Disease*
- *Crop: crop*
- *Location-Karnataka*
- *Symptom: due to warm and humid climate*
- *Urgency: Medium*
- *Advisory: model give multiple ideas to resolve the problem like-scouting, adjust cultivation practices, foliar application.*

3. Key outcomes

- Reduced response time for farmer queries
- Automatic prioritization of critical issues
- Intelligent extraction of agricultural entities
- Cost-efficient AI support system
- Scalable and modular architecture
- Improved support efficiency for agriculture officers

Conclusion-

The Agriculture Support Triage Agent successfully demonstrates how Agentic AI systems can automate agricultural support services. By integrating intent classification, NER, urgency detection, and LLM-based response generation, the system provides intelligent, scalable, and real-time assistance to farmers.

The project enhanced our understanding of:

- Multi-agent AI architecture
- LLM API integration
- Backend API development
- Session and state management
- Caching strategies
- Secure system design

This system can significantly reduce manual workload and improve agricultural productivity through timely intervention.

Future Scope & Enhancements

- Image-based crop disease detection
- Voice query support in regional languages
- Integration with live weather APIs
- Multilingual AI support
- Advanced ML model training using historical data
- Mobile application deployment
- Cloud deployment (AWS / Azure)
- Real-time analytics dashboard for authorities